# Depth-profile extraction and endpoint estimation

## 1 Depth-profile extraction and endpoint estimation

Let $\lambda(\mathbf{r})$ denote the reconstructed activity image, resampled onto a beam-aligned grid with $\mathbf{r} = (x, y, z)$ and the beam (depth) along $z$. The one-dimensional depth–activity profile is the transverse integral over a fixed region of interest $\mathcal{R}$,

$$P(z) \;=\; \int_{\mathcal{R}} \lambda(x, y, z)\, \mathrm{d}x\, \mathrm{d}y \;\approx\; \Delta x\, \Delta y \sum_{(i,j)\in\mathcal{R}} \lambda_{ijk(z)}, \tag{1}$$

where $\mathcal{R} = \{(x, y) : (x - x_c)^2 + (y - y_c)^2 \le \rho^2\}$ is a disc of radius $\rho$ centred on the *beam axis* $(x_c, y_c)$, not on the per-realisation activity centroid: a data-driven centre would inject a stochastic transverse shift into the endpoint and inflate $\sigma_R$. The radius is chosen once, from

$$\rho \;\gtrsim\; n\sqrt{\sigma_{\text{spot}}^2 + \sigma_{\text{PSF}}^2} \;+\; R_{\beta^+}, \qquad n \simeq 3, \tag{2}$$

combining the beam spot size, the reconstruction point-spread width ($\sigma_{\text{PSF}} \approx 1.7\,\text{mm}$ for CRYSP) and the positron range $R_{\beta^+}$. Equation (1) is an integration (a sum), never an average or a normalisation: summing preserves the counting statistics on which the endpoint fit and $\sigma_R$ depend. Because $P(z)$ is a sum of voxel counts it is, to first approximation, Poisson with variance $\mathrm{Var}[P(z)] \approx \bar{P}(z)$; the MLEM reconstruction correlates neighbouring voxels, so this is a working approximation used only to weight the fit, while the authoritative endpoint uncertainty is the ensemble spread defined below.

The activity profile is not a single sigmoid—it rises from the entrance, plateaus, and falls at the distal edge, which lies *upstream* of the Bragg peak because the $\beta^+$-production channels close below their reaction thresholds. The endpoint is therefore estimated within a *fixed* distal analysis window $\mathcal{W} = [z_0^{\text{nom}} - \delta_-,\, z_0^{\text{nom}} + \delta_+]$, bracketing the falloff and defined from the *nominal* range, identically for every arm and realisation; a per-realisation peak search would correlate $\mathcal{W}$ with the very fluctuations being measured. Inside $\mathcal{W}$ the falloff is modelled by a complementary error function,

$$\hat{P}(z) \;=\; b \;+\; a\,\tfrac{1}{2}\,\mathrm{erfc}\!\left(\frac{z - z_0}{\sqrt{2}\,w}\right), \tag{3}$$

so that $z_0$ is by construction the $R_{50}$ point, $a$ and $b$ absorb the plateau height and background (making $z_0$ invariant to the absolute activity scale), and $w$ is the edge width. Any falloff level follows analytically,

$$R_x \;=\; z_0 \;+\; \sqrt{2}\,w\;\mathrm{erfc}^{-1}(2x), \tag{4}$$

with $R_{50}$ primary and $R_{80}, R_{20}$ as secondary tail diagnostics. The parameters are obtained by minimising the Poisson-weighted residual

$$\chi^2 \;=\; \sum_{z\in\mathcal{W}} \frac{\big[P(z) - \hat{P}(z)\big]^2}{\max\!\big(P(z), 1\big)}, \tag{5}$$

which yields both the endpoint $z_0^{(k)}$ and its fit covariance for each realisation $k$.

Finally, over the ensemble of $K$ realisations at a given delivered dose, the mean endpoint estimates the systematic activity-edge position while its spread is the terminal figure of merit,

$$\bar{z}_0 = \frac{1}{K}\sum_k z_0^{(k)}, \qquad \sigma_R = \sqrt{\frac{1}{K-1}\sum_k \big(z_0^{(k)} - \bar{z}_0\big)^2} \;\sim\; \frac{1}{\sqrt{N}}, \tag{6}$$

recovering the counts-limited scaling in the detected-coincidence number $N$. The methodological invariant of the comparison is that the grid, the region of interest $(x_c, y_c, \rho)$, the window $\mathcal{W}$, the model (3) and the estimator (5) are identical across the three geometries: the *only* quantity that differs entering $\sigma_R$ is the reconstructed image content. This is also where the open dual-head geometry is tested most sharply—its limited-angle artefact elongates the activity distribution, and whenever that elongation projects onto the beam ($z$) axis it biases $z_0$ directly, so $\sigma_R$ is reported as a function of beam-axis orientation relative to the head gap.

## 1.1 Implementation and validation

The analysis is realised in two small, reconstruction-agnostic Python modules, so that the only bespoke component is the MLEM reconstruction itself. Module `range_endpoint.py` provides the realisation generator `thin_lm` (step 2, Poisson-thinning of the master list-mode run) and the ensemble aggregator `sigma_R` (step 6). Module `depth_profile.py` provides the profile extractor `depth_profile` (step 4) and a window-aware, Poisson-weighted `fit_endpoint` (step 5) that supersedes the simpler estimator in `range_endpoint.py`. The two are joined at a single seam: within the sweep loop, `depth_profile` and `fit_endpoint` are imported from `depth_profile.py` while `thin_lm` and `sigma_R` are imported from `range_endpoint.py`. Crucially, the fixed analysis window $\mathcal{W}$ of Eq. (3) is constructed once, via `distal_window(`$z_0^{\mathrm{nom}}$`)`, and reused unchanged for every geometry and every realisation—the concrete guarantee that the window is not a per-realisation, noise-correlated quantity.

The correspondence between the formalism and the code is one-to-one, as summarised in Table 1. In particular, the Poisson weighting of Eq. (5) is implemented as a least-squares fit with $\sigma = \sqrt{\max(P, 1)}$ and `absolute_sigma=True`, so that the fit covariance carries the counting-statistics scale; as noted in Section 1, this per-realisation error is used only for the weighting, whereas the reported precision is the ensemble spread $\sigma_R$ of Eq. (6).

Table 1: Correspondence between the formalism and its implementation. Steps refer to the pipeline of Section 1.1; reconstruction (step 3) is external.

| Equation | Step | Function | Module |
|---|---|---|---|
| — | 2 | `thin_lm` | `range_endpoint.py` |
| Eq. (1) | 4 | `depth_profile` | `depth_profile.py` |
| Eq. (2) | 4 | `roi_radius_mm` arg. | `depth_profile.py` |
| $\mathcal{W}$ window | 5 | `distal_window` | `depth_profile.py` |
| Eqs. (3)–(4) | 5 | `fit_endpoint` / `_erfc_model` | `depth_profile.py` |
| Eq. (5) | 5 | `fit_endpoint` (weighted) | `depth_profile.py` |
| Eq. (6) | 6 | `sigma_R` | `range_endpoint.py` |

The extraction chain was validated on a synthetic beam-aligned phantom: a transverse Gaussian beam ($\sigma_\perp \approx 2$ mm) modulated along depth by a plateau-plus-erfc falloff with a known edge at $R_{50} = 150.0$ mm, sampled on a 1.5 mm grid and subjected to Poisson noise. Passing this image through the full profile→window→fit chain recovers $R_{50} = 149.71$ mm with a fit uncertainty of 0.13 mm, confirming sub-voxel, essentially unbiased endpoint localisation at realistic single-realisation statistics.